3.4. ISOTOPIC RATIOS IN NATURE

Mass spectrometric investigations of geologic material has shown that isotopic ratios vary slightly in nature with the largest variations observed for the heaviest and lightest elements. Hence the atomic weight of many elements is a little uncertain, which is indicated be giving fewer decimals than for their isotopes.

For the heaviest elements the reason for the variation in the ratio of isotopes in an element can be related directly to the consequence of naturally occurring radioactive decay. For example, in a thorium mineral the decay sequence terminates in the formation of an isotope of lead with mass number 208. By contrast, in a uranium ore, in which ^{238}U is the most abundant uranium isotope, the primary end product of the decay is an isotope of lead with mass number 206 (see Fig. 1.2). This means that thorium minerals and uranium minerals contain lead with different isotopic composition. In fact, one of the best confirmations of the existence of the radioactive decay series came from the proof that lead had different net atomic weights in different minerals.

Lead from different ores will contain slightly different isotopic abundances due to the fact that all long natural radioactive decay series end up in differet isotopes of lead An unusual use of this fact was made by Andrasko et al. to identify smears and fragments from lead bullets used in a homicide case, so that the suspect could be bound to the case, as the isotopic composition of lead bullets can be identified not only by manufacturer but also by manufacturing date.

The isotopic ratios for the lightest elements depend on in which natural material they are found. The ratio for $^{7}Li/^{6}Li$ varies from 12.20 to 12.48, while that of $^{12}C/^{13}C$ varies from 89.3 to 93.5. The isotopic ratio $^{18}O/^{16}O$ varies by almost 5% as shown in Fig. 3.3. However, since natural oxygen contains only 0.2% ^{18}O, even a 5% variation in the isotopic ratio has very little influence on the atomic weight of oxygen. The variation of natural isotopic ratios for boron and chlorine causes the uncertainties in their abundances shown in Table 3.

3.5. PHYSICOCHEMICAL DIFFERENCES FOR ISOTOPES[2]

Although the isotopic variations in the heaviest elements can be attributed to the consequences of radioactive decay, the variations observed in lighter elements are attributable to chemical behaviour. The rates and equilibria of chemical reactions depend on the masses of the atoms involved in the reactions as is explained in §3.6 and §3.7. As a consequence, isotopes may be expected to have somewhat different quantitative physicochemical values (the *isotope effect*). This effect is naturally more predominant for lighter elements than heavier since for the lighter elements the change in mass by adding a neutron

[2] §§3.5-3.7 outline the scientific basis for isotope effects and isotope separation. This is not essential for the Radiochemistry part of this book.

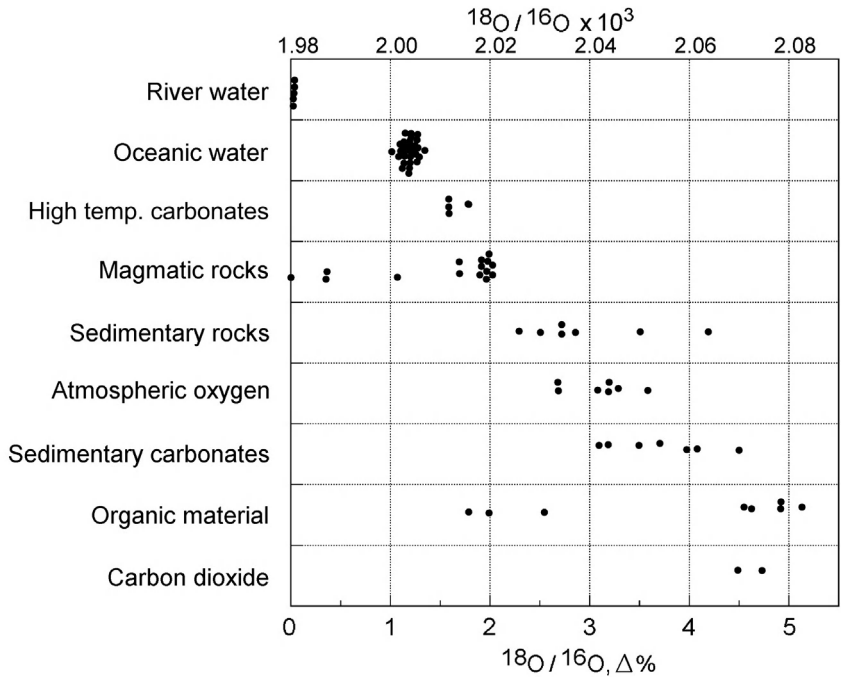


Figure 3.3 Observed $^{18}O/^{16}O$ isotope ratios. (According to Vinogradov.)

may be as high as 100% (for hydrogen) while the change in mass between e.g. ^{235}U and ^{238}U is only about 1%. As examples of the isotope effect, we may note that the freezing point of H_2O is 0°C (273.15 K), while that for heavy water D_2O is 3.82 degrees higher (276.97 K). The boiling point of D_2O is 1.43 K higher than that of H_2O. Similarly, while H_2 boils at 20.26 K, D_2 boils at 23.59 K. As a result of these differences in the boiling points, the vapor pressures at a particular temperature for H_2 and D_2 are not the same and distillation can be used to fractionate hydrogen isotopes. Other physical properties such as density, heat of vaporization, viscosity, surface tension, etc., differ similarly.

The optical emission spectra for isotopes are slightly different since electronic energy levels are dependent on the atomic masses. The light emitted when an electron falls from an outer orbit of main quantum number n_2 to an inner orbit of quantum number n_1 ($n_1 < n_2$) is given by

$$\overline{v} = \mathbf{R}_\infty Z^2 m_{\text{red}} \left(1/n_1^2 - 1/n_2^2\right)/\mathbf{m}_e \tag{3.23}$$

where $\overline{v}$ is the wave number ($1/\lambda$, m^{-1}), $\mathbf{R}_\infty$ is the Rydberg constant (1.097×10^7 m^{-1}), $\mathbf{m}_e$ is the electron (rest) mass, and m_{red} the reduced mass, according to

$$m_{\text{red}}^{-1} = m_e^{-1} + m_{\text{nucl}}^{-1} \tag{3.24}$$

where m_{nucl} is the nuclear mass. For the light hydrogen isotope, the H_α line (i.e. the transition energy between $n_1 = 1$ and $n_2 = 2$) occurs at 656.285 nm, while the corresponding line for the deuterium isotope of hydrogen occurs at 656.106 nm. This difference could be predicted from (3.23) and its observation experimentally in 1932 provided the first evidence for the existence of a heavy hydrogen isotope. This spectral difference has practical importance as it can be used for a spectroscopic analysis of the amount of heavy water in light water. Similar *isotopic line shifts* occur in the spectra of all elements, but are rarely as large as the shift of almost 0.2 nm observed for hydrogen. For the isotopes ^{235}U and ^{238}U, the isotopic shift is 0.025 nm.

3.6. ISOTOPE EFFECTS IN CHEMICAL EQUILIBRIUM

In a mixture of molecules AX and BX, with a common element X, an exchange of the atoms of the common element between the two molecules may occur. When the two compounds have different isotopes X and X*, we may have an *isotope exchange* according to

$$AX + BX^* = AX^* + BX \tag{3.25}$$

The equilibrium constant k in the reaction is given by

$$\Delta G^{\circ} = -\mathbf{R}T \ln k = -\mathbf{R}T \ln \{([AX^*][BX])/([AX][BX^*])\} \tag{3.26}$$

where ΔG° is the (Gibb's) free energy and $\mathbf{R}$ is the molar gas constant. For values of fundamental constants see Appendix III.

3.6.1. The partition function

It has been shown that k deviates slightly from 1, giving rise to the isotopic effects observed in nature for the light elements. This deviation can be calculated by methods of statistical thermodynamics. Only the main features of this theory are given here. The equilibrium constant k can be written

$$k = (F_{AX^*}\, F_{BX})/(F_{AX}\, F_{BX^*}) \tag{3.27}$$

where F is the *grand partition function*, which for each molecule includes all possible energy states of the molecule and the fraction of molecules in each energy state under given external condition. The grand partition function is defined by

$$F = f_{\text{tr}}\, f_{\text{rot}}\, f_{\text{vib}}\, f_{\text{el}}\, f_{\text{nsp}} \tag{3.28}$$

where each term f_j refers to a particular energy form: translation, rotation, vibration, electron movement, and nuclear spin. The two latter will have no influences on the chemical isotope effect, and can therefore be omitted. It can be shown that each separate partition function f_j can be described by the expression

$$f_j = \Sigma g_{j,i} e^{-E\text{j,i}/\mathbf{k}T} \tag{3.29}$$